Heat Transfer to a Fluid in a Stirred Tank

Pre-Lab Plan I

Unit Operations Lab II

01/23/2020

Commented [DL1]: 25.5/35

Group #4

Thursday, PM section

Group Members:

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1. Experimental Objective

The main objective of this experiment is to measure the heat transfer coefficient between the fluid (water for this experiment) and the vessel wall. Additionally, it is required to examine the dependence of the measured heat transfer coefficient on the impeller speed, the presence of baffles, and the fluid properties that are highly dependent on temperature. Finally, the measured coefficients are to be compared to other values that have been officially analyzed and reported by experts.

2. Experimental

2.1. Apparatus

The apparatus for this experiment is displayed in the figure below and its parts are described in the table below.

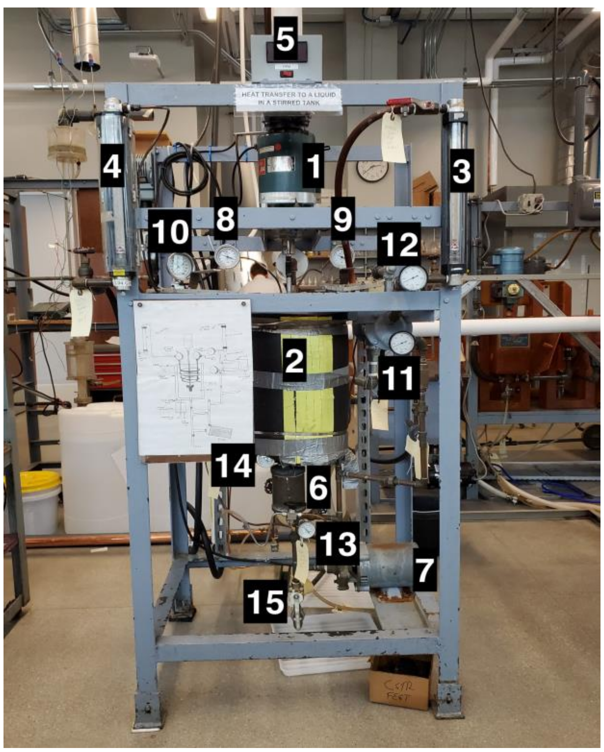

Figure 1 - Process Control System Experiment Setup

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A brief introduction of the apparatus is needed. You may include the flowsheet of the process.

Commented [AZ5]: Figure 1

Commented [AZ6]: Table 1

Table 1 - Apparatus Equipment Explanations

No.	**Equipment Parts**	**Description**
1	**Agitator**	Device that stirs the liquid.
2	**Vessel**	Container that holds the liquid.
3	**Heat Exchanger Return Rotameter**	Manages flowrate of water returning to the vessel.
4	**Cooling Coil Water Rotameter**	Manages flow rate of cooling water circulating through the coil.
5	**Steam In**	Bulk steam in flow to vessel jacket.
6	**Steam Trap**	Condenses steam utilizing cooling water.
7	**Pump**	Circulates water from vessel to heat exchanger.
8	**Cooling Coil Water Return Temperature Gauge**	Reports temperature of cooling water entering the cooling coil.
9	**Cooling Coil Water Exiting Temperature Gauge**	Reports temperature of cooling water exiting the cooling coil.
10	**Vessel Temperature Gauge**	Reports overall temperature of the vessel.
11	**Water Entering Heat exchanger temperature Gauge**	Reports temperature of cooling water entering heat exchanger.
12	**Water Exiting Heat exchanger temperature Gauge**	Reports temperature of cooling water exiting heat exchanger.
13	**Cooling water entering steam trap temperature Gauge**	Reports temperature of cooling water entering steam trap.
14	**Cooling water exiting steam trap temperature Gauge**	Reports temperature of cooling water exiting steam trap.
15	**Vessel Drain**	Relieves vessel of contents to the local sewer main.

Commented [DL7]: Include the steam jacket, baffle and the cooling coil.

Commented [DL8]: Unit and the maximum capacity.

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2.2. Materials and Supplies

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Materials used in this experiment, description and uses are depicted in the table below;

Table 2 - Materials and Supplies Descriptions

Commented [DL18]: Include the scale, graduate cylinder and the stopwatch.

No.	Material	Description
1	**Water**	Medium to conduct heat transfer, cool the steam.
2	**Steam**	Inlet to the system, to measure and calculate the amount of heat transfer, Q.
3	**Air**	Needed to create pressure.
4	**Electricity**	Powers the impeller.

Commented [DL19]: No need for air.

2.3. Experimental Plan

Procedure

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General Operating Procedures

1. Fill tank with water to about 2 inches from the top. Measure height of liquid level.
2. Turn on impeller motor and set to approximately above 600 RPM.
3. Turn on cooling water to the cooling coils, and open control valves fully.
4. Turn on cooling water to the tube-side of the shell and tube heat exchanger.
5. Make sure that the outlet valve in the discharge line of the pump is fully closed and that the suction and bypass valves are fully open.
6. Switch on the pump and allow it to run for a few seconds with the discharge valve closed.
7. Open the discharge valve slowly and close the bypass valve to set a particular recirculation rate. Make sure that the circulating liquid is fed back to the tank.

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8. Slowly open inlet steam valve (11/2 turns) after cooling water is supplied to the steam condenser
9. At steady state, record all relevant properties namely, flow rates, temperatures, speed of impeller etc.

Part A (Measurements):

1. $h_{i,wall}$. At constant steam flow-rate to the vessel, low impeller speed (approximately 600 RPM), with water flowing through the cooling coils and with the heat exchanger flow-rate such that the temperature in the vessel $t_3$is about65^0C. Try to keep the vessel temperature below 85^0Cin most of your experiments. Record the measurements needed to calculateQ_{13}.
2. $h_{i,wall}$as a function of impeller speed. Repeat the above measurements at two additional higher impeller speeds.
3. $h_{i,wall}$as a function of temperature. Repeat the measurements for two different heat exchanger flow-rates so that two different vessel temperatures are obtained. Use your second impeller speed settings.

Part B (Measurements)

1. Measurements will be made at approximately the same conditions as in part A, except with baffles inserted.
2. After steady state has been reached in each case of experiment A, carefully lift the vessel cover plate and slip the baffles into place, making sure the top flat section is seated over several allen screw heads.
3. After the new steady state has been reached, take all necessary measurements.

Commented [DL22]: What's the criteria of the steady state?

Commented [DL23]: List all parameters need to be measured.

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Commented [DL27]: Be specific.

Project Safety Evaluation (10 pts)

Project Title: **Heat Transfer to a Fluid in a Stirred Tank**

Date: **January 23, 2020**

Hazard	*Yes/No*	Description
Potential electrical hazards?	Yes	Impeller operates on electricity which can cause electric shocks.
Potential mechanical hazards?	Yes	The impeller turning at a fast speed, if something ~~get~~gets caught in it, it can be damaged
Potential pressure hazards?	Yes	Depressurization can occur if pressure valve is not opened and closed at specified points.
Potential temperature hazards?	No	Experiment will be conducted at room temperature.
Hazardous/toxic chemicals involved?	Yes	CARBOPOL should not be inhaled and it should not come in contact with the skin. If this occurs, wash skin in contact thoroughly and seek medical attention as needed.
Gloves required?	Yes	Heat resistant gloves may be required when touching hot objects.
MSDS reviewed by all team members?	Yes	Sand and Ion exchange Resin MSDS sheet reviewed.
Process Parameters	*Max*	Possible projectile hazard if pressure vessel is over pressurized and fluid is ejected from the pressure vessel.

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Commented [AZ29]: It looks like you copied parts of this fromyour Viscosity experiment from last semester.

Commented [AZ30]: Pump is below vessel with potential water leaking onto pump.

Commented [AZ31]: yes

Commented [AZ32]: Pressure of what, water or steam?

Commented [DL33]: The CST is hot.

Commented [DL34]: ?
No.

Commented [DL35]: ?
CAS# for water.

	Expected	
Temperature (ºC)	25℃	Air, Sand, and Resin also at room temperature.
Pressure (psig)	40 psig	Inlet air stream will operate at 40 psig.
Safety Controls	Yes	Make sure to drain the tanks and turn the computer and controllers off when finished with the lab.

Commented [DL36]: ?
120 degree Celsius for the steam.

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14.7 psia.

Commented [DL38]: ?
Shut-off valves are provided.

I have read relevant background material for the Unit Operations Laboratory entitled: "Heat Transfer to a Fluid in a Stirred Tank" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

01/23/2020 _____ Javier Eduardo Mendieta Castillo

signature of team member

date

Omar Zahdan ________________

________ 01/23/2020 _______

signature of team member

date

Omar Takrouri __________________

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signature of team member

date

Joel Finlon ____________________

________ 01/23/2020 _______________

Heat Transfer to a Fluid in a Stirred Tank **University of Illinois at Chicago**

signature of team member

date

01/23/2020 ____ Yusuf Sorunke

date signature of team member